# Smart Complaint Management System

# Software Test Report — SCMS_Test_Report_v1.0

## Integration Testing: TC-01 to TC-12

**Test scope:** Registration, authentication, complaint submission, authorization, administration, security validation, response-time verification, and automated regression testing.

**Test environments:** Local Flask application and deployed Render instance.

**Automated regression result:** 101 passed, 0 failed. The suite also includes 70 login-lockout tests, all of which passed.

## 1. Test Case Results

| ID | Test case | Expected result | Actual result / evidence | Result |
|---|---|---|---|---|
| TC-01 | Registration | Valid user is registered and redirected to login. | Automated test test_tc01_registration; registration flow completed with redirect to /login. | PASS |
| TC-02 | Duplicate registration | Existing email is rejected. | Automated test test_tc02_duplicate_registration confirms duplicate email validation. | PASS |
| TC-03 | Valid login | Valid credentials authenticate the user and redirect to dashboard. | Automated test test_tc03_valid_login; redirect observed to /dashboard. | PASS |
| TC-04 | Invalid login | Invalid credentials are rejected with a generic error. | Automated test test_tc04_invalid_login; generic invalid-credentials message verified. | PASS |
| TC-05 | Complaint submission | Authenticated user can submit a valid complaint. | Automated test test_tc05_complaint_submission; successful submission redirects to /complaints. | PASS |
| TC-06 | Empty complaint | Missing required complaint fields are rejected. | Automated test test_tc06_empty_complaint_rejected verifies required-field validation. | PASS |
| TC-07 | User complaints | User can view their own complaint list. | Automated test test_tc07_user_complaints verifies /complaints and submitted complaint visibility. | PASS |
| TC-08 | Admin access | Admin can access the admin dashboard. | Automated test test_tc08_admin_access verifies redirect to /admin. | PASS |
| TC-09 | Unauthorized admin access | Normal user is denied admin access. | Automated test test_tc09_normal_user_denied_admin verifies HTTP 403. | PASS |
| TC-10 | Admin dashboard | Authorized admin can view dashboard. | Automated test test_tc10_admin_can_view_dashboard verifies HTTP 200 and dashboard content. | PASS |
| TC-11 | Response time | Application responses meet the project response-time target. | Measured deployed response times: / 0.671 s, /register 0.873 s, /login 0.908 s; all under 2 seconds for sampled requests. | PASS |
| TC-12 | Security / credential protection | Malformed/malicious inputs do not bypass controls or execute as code; secrets are protected. | Verified: 1000+ character input handled without 500; XSS payload did not execute; SQL-injection-style login did not bypass authentication. Automated regression suite: 101 passed. | PASS |

## 2. Automated Testing Evidence

The complete pytest suite was executed after the complaint-detail tests and test-expectation updates. Final result: **101 passed, 0 failed**. The dedicated login-lockout suite also completed with **70 passed, 0 failed**. The remaining pytest output consisted only of deprecation warnings.

## 3. Defects and Retesting

During integration testing, three stale test expectations were identified: registration, valid login, and complaint submission expected HTTP 200 while the implemented application correctly returns redirects (HTTP 302). The tests were updated to assert the implemented redirect behavior. A subsequent full run reported 100 passed and 1 failed because the valid-login redirect target was /dashboard rather than /complaints. That expectation was corrected, followed by the final full run with 101/101 tests passing.

Issue #92 complaint-detail access-control tests were also added and passed 3/3: complaint owner receives 200, another user receives 404, and a logged-out user is redirected to /login.

## 4. Security Test Evidence

Security validation included an oversized complaint description (>1000 characters), an XSS payload (<script>alert('XSS')</script>), and a SQL-injection-style login input (' OR 1=1 --). No server error, script execution, or authentication bypass was observed. Login-lockout regression testing additionally passed all 70 automated cases.

## 5. Conclusion

**Overall result: PASS.** All twelve test cases have recorded results, automated regression testing completed with 101/101 tests passing, and identified test defects were corrected and retested successfully. The report is suitable as the Sprint 2 integration-test deliverable for Issue #53.